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AGENDA

1. Why Machine Learning.
2. Types of Machine Learning.
3. How to approach solving a business problem.
4. supervised learning overview
5. Unsupervised learning overview.
6. Reinforcement learning overview.

CHAPTER-1 ML FOUNDATIONS

1. Why MACHINE LEARNING?

When we have statistics, why ML?

Statistics purpose
↳ Insights + Decision
↳ Human Decides

ML purpose
↳ Insights + Prediction.
↳ Machine Decides.

2 TYPES OF MACHINE LEARNING:-

* Supervised learning
* Unsupervised learning
* Reinforcement learning

* Self-supervised learning
* Semi-supervised learning.

What is Machine Learning:- { Insights + Prediction }

Eg:-

* Gmail Spam filter → You didn't tell Gmail what "spam" looks like, it learned.
* YouTube Recommendation → You didn't tell youtube what "video" will you watch next, it learned.

WHY MACHINE LEARNING:-

Problem: Manual rules break down in complex problems.

Eg: Billions of emails / videos / transcation can't be handled by manually.

Solution:- Build a system, that automatically does this job and scale it.

Benefits:- Saves cost, improves decisions.

Types of Machine Learning:-

1. Supervised Learning → X & Y present in the Data
2. Unsupervised Learning → only X present
3. Reinforcement Learning → Like training a kid with rewards system.

How does Machine Learning work? (Pipeline).

1. Business understanding
2. Data collection.
3. Data understanding
4. Data preparation
5. Model Building.
6. Model Training
7. Model Testing
8. Model Evaluation
9. Model Deployment
10. CICD [Continuous Integration / Continuous Development]

UNSUPERVISED LEARNING:-

i) What is:- It is way to teach machines to find patterns or groups in data without knowing the answers beforehand. Unlike supervised learning, there are no output labels (Y) available.

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(ii) When to use it

We use unsupervised learning when we only have input data (x) and we want to discover hidden structures, clusters or patterns in it.

(iii) Where to use it? Eg:-

Imagine a retail store wants to understand its customers better. They have data like Age, Monthly spend, no. of visits per month. But they don't know which customers are high spenders or low spenders.

Manually, its difficult to identify patterns. With unsupervised learning, the machine can automatically group similar customers together.

(iv) Dataset Requirement.

We need input data only. For eg: 15 customers will features like age, & monthly spend. Theres no label saying high spender or low spender - The machine will discover that.

customer_ID	Age	Monthly spend
C001	22	3000
C002	45	9000
C003	29	3500
C004	33	4000
C005	50	12000
...	...	...
C015	42	10000

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REINFORCEMENT LEARNING:-

What is: It is a way of teaching machines how to make decisions by giving them rewards (for good actions) and penalties (for bad ones). The machine then learns which choices give maximum reward in the long run.

2. When to use it:
"we use reinforcement learning, when the outcome depends on a sequence of decisions, and the machine has to learn by trail-and-error"
* No ready-made input-output dataset like supervised learning.
* Instead, the machine interacts with an environment and learns strategy over time.

3. Where to use it? Eg:-
Think of a child learning to play a video games. The first time, they just press random buttons. Slowly, they learn which actions make them win points and which cause them to "lose a life". They eventually master the strategy.

4. Data Requirement :-
⇒ Unlike supervised learning, we don't give X & Y columns ahead of time. Instead, we define:
* Environment → where the agent operates.
* Agent → The learner / decision-maker
* Actions → what choices are available
* Reward → feedback signal (positive or negative).
* Goal → maximize long-term rewards.